A side check: MMD-LOCO against vanilla MMD on a covariate shift

This note is not the FSDS horse-race. It only asks whether the leave-one-coordinate MMD scores, turned into a single permutation statistic, detect a mean shift in X more often than the joint MMD on the same kernel.

The kernel is $\exp(-\gamma\|x-y\|^2)$ with γ equal to one over the median pooled squared distance. γ is computed once on the observed pool and held fixed. Dropping coordinate j subtracts that coordinate's squared differences from the three distance matrices, which is the joint kernel on the remaining coordinates at the same γ. The vector of interest is $\mathrm{MMD}^2 - \mathrm{MMD}^2_{-j}$. Two scalars are tested: the maximum coordinate, and the sum of the positive coordinates. The baseline is the unbiased MMD^2 itself. A normalized version, $\mathrm{NAMMD}^2 = \mathrm{MMD}^2_u / \sqrt{\bar{k}_{xx}\bar{k}_{yy}}$, is carried along because the same drop applies to it. Each test uses the same label permutations. The p-value is $(1 + \#\{T_b \geq T\})/(B+1)$.

Each batch has $n = 120$ reference rows and n new rows. Coordinates are standard normal, independent in the main grid, with an AR(1) check beside it. The new batch adds a constant to the first s coordinates. $B = 199$ permutations and 80 Monte Carlo replicates. The dashed line in the figure is $\alpha = 0.05$. ECMMD is omitted: it needs a response and is a conditional discrepancy, not a test of $P(X)$.

On one shifted coordinate the maximum is the more sensitive of the two. At $p = 50$ and shift 0.60, max LOCO rejects in 0.95 of the 80 replicates and vanilla MMD in 0.65; 25 replicates are rejected only by the maximum and 1 only by MMD. At $p = 20$ and shift 0.50 the rates are 0.85 and 0.60 (22 only by the maximum, 2 only by MMD). A shift in every coordinate reverses this. At $p = 50$ and shift 0.10, vanilla MMD rejects in 0.81 and the maximum in 0.19 (50 replicates only by MMD). The sum of the positive coordinates stays nearer the joint statistic on that dense shift (0.65) than the maximum does. Null rejection across the grid sits between 0 and 0.11.

Table 1: Independent coordinates. Rejection rate at $\alpha = 0.05$. Each row is one covariate-shift cell. The six statistics share the kernel, the bandwidth, and the permutations.

p	s, shift	MMD	max LOCO	positive-sum LOCO	NAMMD	max NAMMD-LOCO	positive-sum NAMMD-LOCO
20	1, 0.00	0.04	0.07	0.07	0.04	0.06	0.06
20	1, 0.20	0.16	0.12	0.23	0.16	0.14	0.20
20	1, 0.35	0.33	0.46	0.33	0.33	0.44	0.31
20	1, 0.50	0.60	0.85	0.72	0.60	0.84	0.68
20	1, 0.80	0.99	1.00	0.99	0.99	1.00	0.99
20	4, 0.00	0.05	0.11	0.10	0.05	0.10	0.05
20	4, 0.20	0.39	0.21	0.34	0.39	0.24	0.35
20	4, 0.35	0.93	0.80	0.89	0.93	0.84	0.93
20	4, 0.50	1.00	0.97	1.00	1.00	0.99	1.00
20	4, 0.75	1.00	1.00	1.00	1.00	1.00	1.00
20	20, 0.00	0.05	0.03	0.05	0.05	0.03	0.06
20	20, 0.10	0.38	0.12	0.29	0.38	0.14	0.35
20	20, 0.20	1.00	0.49	1.00	1.00	0.72	1.00
20	20, 0.35	1.00	0.95	1.00	1.00	1.00	1.00
20	20, 0.50	1.00	0.99	1.00	1.00	1.00	1.00
50	1, 0.00	0.06	0.03	0.06	0.06	0.03	0.09
50	1, 0.40	0.30	0.54	0.36	0.30	0.54	0.30
50	1, 0.60	0.65	0.95	0.80	0.65	0.95	0.74
50	1, 0.85	0.94	1.00	0.97	0.94	1.00	0.94
50	1, 1.15	1.00	1.00	1.00	1.00	1.00	1.00
50	4, 0.00	0.04	0.00	0.04	0.04	0.00	0.04
50	4, 0.30	0.57	0.49	0.59	0.57	0.53	0.61
50	4, 0.50	0.99	0.99	1.00	0.99	1.00	1.00
50	4, 0.75	1.00	1.00	1.00	1.00	1.00	1.00
50	4, 1.00	1.00	1.00	1.00	1.00	1.00	1.00
50	50, 0.00	0.04	0.05	0.06	0.04	0.05	0.05
50	50, 0.10	0.81	0.19	0.65	0.81	0.24	0.76
50	50, 0.18	1.00	0.54	1.00	1.00	0.74	1.00
50	50, 0.30	1.00	0.85	1.00	1.00	1.00	1.00
50	50, 0.45	1.00	1.00	1.00	1.00	1.00	1.00

Table 2: AR(1) correlation 0.5. Rejection rate at $\alpha = 0.05$. Each row is one covariate-shift cell. The six statistics share the kernel, the bandwidth, and the permutations.

p	s, shift	MMD	max LOCO	positive-sum LOCO	NAMMD	max NAMMD-LOCO	positive-sum NAMMD-LOCO
20	1, 0.00	0.05	0.03	0.03	0.05	0.03	0.05
20	1, 0.35	0.25	0.44	0.29	0.25	0.44	0.26
20	1, 0.60	0.61	0.97	0.78	0.61	0.96	0.75
20	20, 0.00	0.05	0.03	0.04	0.05	0.03	0.06
20	20, 0.20	0.91	0.42	0.69	0.91	0.56	0.84
20	20, 0.40	1.00	0.86	0.99	1.00	1.00	1.00

Table 3: Paired comparison with vanilla MMD on the same replicates. "Only max" counts replicates rejected by max LOCO and not by MMD. "MMD, not max" is the reverse. The last two columns are the same split for the sum of positive LOCO scores.

p	s, shift	MMD	max LOCO	only max	MMD, not max	pos. sum	only pos.	MMD, not pos.
20	1, 0.00	0.04	0.07	5	2	0.07	4	1
20	1, 0.20	0.16	0.12	5	8	0.23	7	2
20	1, 0.35	0.33	0.46	16	5	0.33	5	5
20	1, 0.50	0.60	0.85	22	2	0.72	12	2
20	1, 0.80	0.99	1.00	1	0	0.99	0	0
20	4, 0.00	0.05	0.11	6	1	0.10	4	0
20	4, 0.20	0.39	0.21	5	19	0.34	5	9
20	4, 0.35	0.93	0.80	1	11	0.89	1	4
20	4, 0.50	1.00	0.97	0	2	1.00	0	0
20	4, 0.75	1.00	1.00	0	0	1.00	0	0
20	20, 0.00	0.05	0.03	1	3	0.05	1	1
20	20, 0.10	0.38	0.12	4	24	0.29	2	9
20	20, 0.20	1.00	0.49	0	41	1.00	0	0
20	20, 0.35	1.00	0.95	0	4	1.00	0	0
20	20, 0.50	1.00	0.99	0	1	1.00	0	0
50	1, 0.00	0.06	0.03	2	5	0.06	3	3
50	1, 0.40	0.30	0.54	22	3	0.36	8	3
50	1, 0.60	0.65	0.95	25	1	0.80	14	2
50	1, 0.85	0.94	1.00	5	0	0.97	3	0
50	1, 1.15	1.00	1.00	0	0	1.00	0	0
50	4, 0.00	0.04	0.00	0	3	0.04	2	2
50	4, 0.30	0.57	0.49	12	19	0.59	5	4
50	4, 0.50	0.99	0.99	1	1	1.00	1	0
50	4, 0.75	1.00	1.00	0	0	1.00	0	0
50	4, 1.00	1.00	1.00	0	0	1.00	0	0
50	50, 0.00	0.04	0.05	4	3	0.06	3	1
50	50, 0.10	0.81	0.19	0	50	0.65	2	15
50	50, 0.18	1.00	0.54	0	37	1.00	0	0
50	50, 0.30	1.00	0.85	0	12	1.00	0	0
50	50, 0.45	1.00	1.00	0	0	1.00	0	0
20	1, 0.00, $\rho = 0.5$	0.05	0.03	2	4	0.03	0	2
20	1, 0.35, $\rho = 0.5$	0.25	0.44	16	1	0.29	3	0
20	1, 0.60, $\rho = 0.5$	0.61	0.97	29	0	0.78	15	2
20	20, 0.00, $\rho = 0.5$	0.05	0.03	1	3	0.04	1	2
20	20, 0.20, $\rho = 0.5$	0.91	0.42	0	39	0.69	0	18
20	20, 0.40, $\rho = 0.5$	1.00	0.86	0	11	0.99	0	1

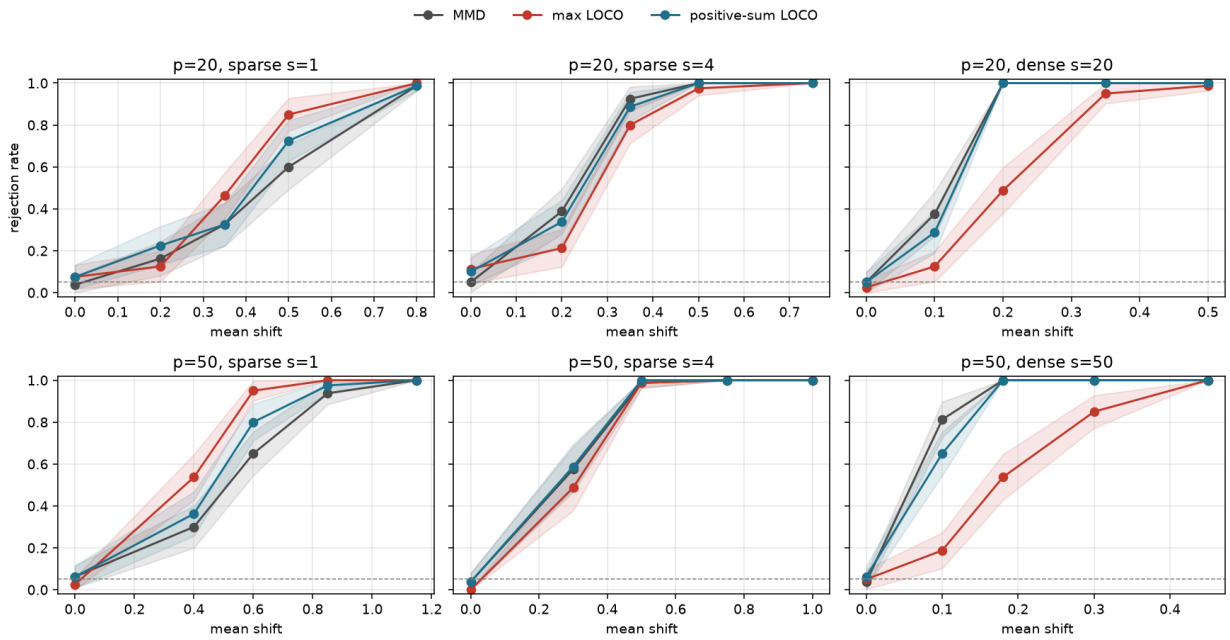


Figure 1: Independent-coordinate grid. Curves are rejection rates. Bands are pointwise 1.96 Monte Carlo standard errors. The horizontal line is 0.05.